

MA.912.GR.2.6

Benchmark

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

Benchmark Clarifications:

Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.2
- MA.912.GR.1.6

Terms from the K-12 Glossary

- Coordinate Plane
- Origin
- Reflection
- Rigid Transformation
- Rotation
- Translation

Vertical Alignment

Previous Benchmarks

- MA.8.GR.2.1, MA.8.GR.2.2, MA.8.GR.2.3

Next Benchmarks

Purpose and Instructional Strategies

In grade 8, students identified a single transformation given the preimage and the image and learned that when the transformation is a reflection, a rotation or a translation, those transformations preserve congruence, that is, the preimage is mapped onto a copy of itself. In Geometry, students determine whether two figures are congruent and justify their answers using a sequence of rigid motions. This leads to the definition of congruence in terms of rigid transformations. (MTR.5.1)

- When identifying the transformations in the sequence, specify the vertical and horizontal translations, center and angle of the rotation, clockwise or counterclockwise, line of reflection, when needed.
- Instruction includes describing the rigid transformations using words and using coordinates.
- It is important to identify corresponding parts between the preimage and the image leading to the congruence statement and the congruence of the corresponding parts (angles and sides).
- Instruction includes using examples to compare transformations preserving angle measures and distance versus transformations just preserving angle measures. Include situations where the preimage and the image are not congruent to show how rigid motions will fail mapping one figure onto the other. (MTR.4.1)

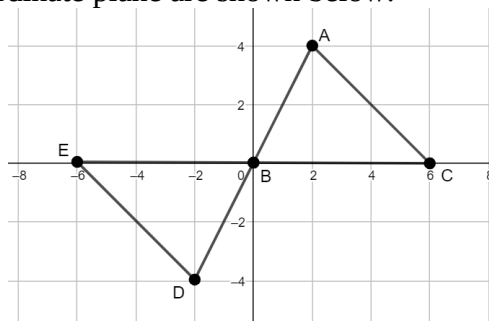
Common Misconceptions or Errors

- Students may have trouble seeing congruence when a reflection is needed because they limit their thinking to sliding a figure around without turning it over.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.4.1)

Two triangles on the coordinate plane are shown below.



Part A. What transformation(s) could be applied to map triangle EBD onto triangle CBA ?

Part B. Once the transformation is completed, how can you determine if the two triangles are congruent?

Instructional Items

Instructional Item 1

Describe the sequence of transformations that could be used to prove that the two quadrilaterals shown are congruent.



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*